Kazi Samin Mubasshir

PH.D. CANDIDATE, DEPARTMENT OF COMPUTER SCIENCE, PURDUE UNIVERSITY

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Research Interests

Cellular and NextG Security (LTE/5G, O-RAN, Voice over WiFi) | Protocol Security Testing and Specification Analysis | ML-Based Network Intrusion Detection and Explainable Mitigation | LLM-Powered Cloud Security Analytics, Retrieval-Augmented Investigation and Agentic Triage

Education

Purdue University *West Lafayette, IN*

Ph.D. in Computer Science *Aug 2022 - Jul 2027 (expected)*

- **Thesis:** Resilient Cellular Networks: Multi-Layered Defense Against Persistent Threats
- **Advisor**: Dr. Elisa Bertino

Bangladesh University of Engineering and Technology *Dhaka, Bangladesh*

B.Sc. in Computer Science and Engineering *Feb 2016 - Feb 2021*

- **Thesis:** Aligner Ensembling, Batch Filtering, and New Datasets for Bengali-English Machine Translation
- **Advisor**: Dr. Rifat Shahriyar · Graduated with Distinction, CGPA 3.81/4.00

Work Experience

Amazon Web Services (AWS)

Applied Scientist Intern *Summers 2024 – 2026*

Security, Search and Observability · New York, NY *Summer 2026*

- Developing agentic AI methods that automate security triage workflows over large-scale search and observability data, reducing manual analyst effort.

Security Analytics and AI Research (SAAR) · Boston, MA *Summer 2025*

- Designed an LLM-based approach for automatically identifying high-severity vulnerability classes in cloud environments without requiring labeled training data, achieving high accuracy and consistency in evaluations.

Security Analytics and AI Research (SAAR), External Security Services (ESS) · Boston, MA *Summer 2024*

- Built an LLM-based research prototype that augments security investigations with relevant context retrieved in real time from cloud data sources.

Purdue University *West Lafayette, IN*

Graduate Research Assistant · Cyber Space Security Lab (Cyber2Slab) *Fall 2022 - Present*

- Built FBSDetector, an ML system that detects fake base stations and the multi-step attacks they launch directly from LTE/5G network traces, by curating a large labeled trace dataset and engineering stateful protocol features; large-scale evaluations and open-source artifacts show high-accuracy, low-false-positive detection on commodity devices (USENIX Security 2025, joint first author).
- Contributed to VWAttacker, a systematic security-testing framework for Voice-over-WiFi user equipment that uncovered downgrade, information-leakage, and denial-of-service weaknesses across commercial devices (IEEE INFOCOM 2026).
- Co-authored work that turns feature-attribution explanations of 5G anomaly-detection models into concrete, actionable mitigation steps for analysts (IEEE ICNC 2026).
- Co-created SPEC5G, the first public NLP dataset for 5G protocol analysis (3.5M+ sentences from specification documents), enabling automated security text classification and summarization (IJCNLP-AACL 2023).

Graduate Teaching Assistant · CS 180: Problem Solving and Object-Oriented Programming *Spring 2025, Fall 2025*

- Led labs, graded projects, and held weekly office hours for a $\sim$900-student course.

Advanced Chemical Industries (ACI) Ltd. *Dhaka, Bangladesh*

Machine Learning Engineer *Fall 2020 - Summer 2022*

- Shipped customer-purchase prediction models (lapsed-customer and next-purchase-amount) that guided retail business decisions, collaborating with a ten-person MIS analytics team.
- Built a speech-to-sentiment pipeline for the ACI call center, automating customer sentiment analysis and FAQ mining.
- Forecast Yamaha motorbike warranty claims to guide inventory planning, showing results in a business dashboard using Python and Django as backend.

Bangladesh University of Engineering and Technology *Dhaka, Bangladesh*

Undergraduate Research Assistant *Spring 2019 - Fall 2020*

- Helped build the largest Bengali-English parallel corpus to date via novel aligner-ensembling and batch-filtering techniques, achieving substantial BLEU improvements over prior translation systems (EMNLP 2020).
- Co-authored BanglaBERT, a Bangla language model that set a new state of the art on Bangla language-understanding benchmarks (NAACL 2022), and XL-Sum, a 1M-article abstractive summarization dataset covering 44 languages (ACL-IJCNLP 2021).

Publications

- Karim, Imtiaz, Hyunwoo Lee, Hassan Asghar, **Kazi Samin Mubasshir**, Seulgi Han, Mashroor Hasan Bhuiyan, and Elisa Bertino. "VWAttacker: A Systematic Security Testing Framework for Voice over WiFi User Equipments." IEEE INFOCOM 2026.
- Uccello, Federica, Imtiaz Karim, **Kazi Samin Mubasshir**, Elisa Bertino, and Simin Nadjm-Tehrani. "Mission Explainable: From Feature Attribution to Mitigation in 5G Anomaly Detection." IEEE ICNC 2026.
- **Kazi Samin Mubasshir***, Imtiaz Karim*, Elisa Bertino, Gotta Detect 'Em All: Fake Base Station and Multi-Step Attack Detection in Cellular Networks, 34th USENIX Security Symposium, 2025 (*joint first authors).
- Hassan Asghar, MH Hwang, J Joo, H Kang, YJ Kwon, **Kazi Samin Mubasshir**, Imtiaz Karim, et al., Poster: Automated Security Property Extraction from Protocol Specifications, IEEE ICNP 2025.
- Imtiaz Karim, **Kazi Samin Mubasshir**, Mirza Masfiqur Rahman, Elisa Bertino, SPEC5G: A Dataset for 5G Cellular Network Protocol Analysis, Findings of IJCNLP-AACL 2023.
- Abhik Bhattacharjee, Tahmid Hasan, **Kazi Samin**, Md Saiful Islam, M. Sohel Rahman, Anindya Iqbal, Rifat Shahriyar, BanglaBERT: Combating Embedding Barrier for Low-Resource Language Understanding, Findings of NAACL 2022.
- Tahmid Hasan, Abhik Bhattacharjee, Md Saiful Islam, **Kazi Samin**, Yuan-Fang Li, Yong-Bin Kang, M. Sohel Rahman, Rifat Shahriyar, XL-Sum: Large-Scale Multilingual Abstractive Summarization for 44 Languages, Findings of ACL-IJCNLP 2021.
- Tahmid Hasan, Abhik Bhattacharjee, **Kazi Samin**, Masum Hasan, Madhusudan Basak, M. Sohel Rahman and Rifat Shahriyar, Not Low-Resource Anymore: Aligner Ensembling, Batch Filtering, and New Datasets for Bengali-English Machine Translation, EMNLP 2020.

Academic Service

- **Organizing:** Web Chair, NextG Networks Cryptography and Security Workshop (co-located with ACNS 2026)
- **Program Committee:** USENIX Security 2027; RAID 2026 (Associate PC); NextG-Sec 2026
- **Reviewer:** IEEE Communications Magazine; IEEE Trans. on Mobile Computing; IEEE Trans. on Networking
- **Sub-Reviewer:** ACM TOPS; IEEE S&P; CCS; USENIX Sec; NDSS; Euro S&P; ACNS; RAID; ICDCS; CODASPY (2023–2026)
- **Artifact Evaluation Committee:** CCS ('24,'25); USENIX Sec '23; WOOT @ IEEE S&P '23
- **Mentoring:** CSGSB Mentor-Mentee Program ('24,'25), Mentored 6 Purdue CS graduate students

Media Coverage and Software Artifacts

- **SPEC5G**: github.com/SysNetS/SPEC5G · **FBS-Detector**: github.com/SysNetS/fbsdetector
- **FBSDetector in the news:**
  - Researchers uncover fake base stations in cellular networks using machine learning (Purdue Computer Science)
  - Fake Base Station Detection, Research project conducted on the POWDER platform (Platforms for Advanced Wireless Research)

Talks and Presentations

- Cellular Network Security, invited lecture, CS 6301 (Security of Emerging Networks), UT Dallas, Spring 2026
- Gotta Detect 'Em All: Fake Base Station and Multi-Step Attack Detection, USENIX Security '25, Seattle, Aug 2025.
- Detecting Fake Base Stations in Cellular Networks, AWS-SAAR Tech Talks, Amazon, Boston, Jul 2024.
- Fake Base Station Detection on PAWR/POWDER platforms: Young Gladiators-2023, Northeastern University, Jun 2023; MERIF-2023, Boston University, May 2023.

Awards and Achievements

2025	**Travel Awards & Student Grant,** PGSG Travel Award; Purdue CoS Graduate Travel Award; USENIX Security Symposium Student Grant
2022	**AI Competitions,** Robi Datathon 2.0 (runners-up); AI for Bangla 1.0 (2nd runners-up)
2016-21	**University Technical Scholarship,** Bangladesh University of Engineering and Technology (BUET)
2017-20	**Dean's List,** All four years of undergraduate studies

Skills

- **Programming**: Python, C/C++, Java and SQL
- **Machine Learning & Explainability**: Deep Learning, Unsupervised Classification, Anomaly Detection, Feature Engineering, Model Quantization, Feature Attribution and Explanation-Guided Decision Support
- **Large Language Model Systems**: Retrieval-Augmented Generation, Instruction Tuning, Parameter-Efficient Fine-Tuning (LoRA), LLM Agents, Tool Use, Multimodal LLMs and LLM Evaluation
- **NLP & Data-Centric AI**: Text Classification, Abstractive Summarization, Machine Translation, Natural Language Inference, Question Answering, Sequence Labeling, Structured Information Extraction, Text-to-SQL, Dataset and Benchmark Curation, Heuristic Design
- **Cellular & Network Security**: LTE/5G Protocol Analysis, Voice over Wi-Fi Security, O-RAN Security, 3GPP Specification Analysis, Stateful Network-Trace Analysis, Threat Modeling and Security Testing
- **Cloud Security Analytics**: Search and Observability, Vulnerability Analysis, Security Investigation Automation and Agentic Security Triage